

Answer Key & Marking Scheme

Subject: Mathematics | **Test:** Test 3 (Day 21, 30 May 2026) — Weekly
Maximum Marks: 80

Syllabus: Real Numbers • Polynomials • Pair of Linear Equations in Two Variables • Quadratic Equations • Arithmetic Progressions • Triangles (cycle revision)

SECTION A — Answer Key (1 mark each)

- Q1. (b) 18 — $144 = 2 \cdot 3^2$, $198 = 2 \cdot 3^2 \cdot 11$, common = $2 \cdot 3^2 = 18$.
- Q2. (c) 60 — $\text{LCM}(12, 15) = 60$.
- Q3. (b) 1 — a parabola crossing x-axis at one point has 1 zero (touches axis).
- Q4. (a) $4/3$ — $\alpha + \beta = 4$, $\alpha\beta = 3$. $1/\alpha + 1/\beta = (\alpha + \beta)/\alpha\beta = 4/3$.
- Q5. (b) $15/4$ — $a \cdot \frac{1}{a} = b \cdot \frac{1}{b} \Rightarrow 3/2 = 2k/5 \Rightarrow k = 15/4$.
- Q6. (b) x-axis.
- Q7. (c) no real roots — $D = -4 < 0$.
- Q8. (b) $b + c = 0$ — $-b/a = c/a \Rightarrow b + c = 0$.
- Q9. (a) 84 — $a + 16d = 4 + 80 = 84$.
- Q10. (a) $n(n+1)/2$.
- Q11. (c) 1, 3, 5, 7, ... — constant difference 2.
- Q12. (b) square of ratio of corresponding sides.
- Q13. (c) 12 cm — $PR/AC = PQ/AB \Rightarrow PR = 6 \cdot 8/4 = 12$.
- Q14. (b) 26 m — $\sqrt{(10^2 + 24^2)} = \sqrt{676} = 26$.
- Q15. (a) 30° .
- Q16. (a) terminating — $8 = 2^3$, denominator of form $2^m \cdot 5^n$.
- Q17. No real roots (complex/imaginary roots).
- Q18. $a = 7 + (n-1) \cdot 6 = 205 \Rightarrow n - 1 = 33 \Rightarrow n = 34$.
- Q19. $\angle F = \angle C = 180^\circ - 50^\circ - 60^\circ = 70^\circ$.
- Q20. $156 = 2^2 \times 3 \times 13$.

SECTION B — Answer Key (2 marks each)

- Q21. Suppose $5 - \sqrt{3}$ is rational, say p/q ($q \neq 0$). Then $\sqrt{3} = 5 - p/q = (5q - p)/q$, which is rational — contradicting the given fact that $\sqrt{3}$ is irrational. Hence $5 - \sqrt{3}$ is irrational. (2)
- Q22. $x^2 + 7x + 10 = (x + 5)(x + 2)$. Zeroes: **-5, -2**. Sum = $-7 = -(\text{coeff of } x)/\text{coeff of } x^2$. ✓ Product = $10 = \text{constant term}/\text{coeff of } x^2$. ✓ (2)
- Q23. From $x - y = -1$, $x = y - 1$. Substituting: $3(y - 1) - 5y = -1 \Rightarrow -2y - 3 = -1 \Rightarrow y = -1$. Then $x = -2$. **x = -2, y = -1**. (2)
- Q24. $D = b^2 - 4ac = 25 - 24 = 1 > 0$. **Roots are real and distinct (and rational since D is a perfect square)**. (2)
- Q25. $AC^2 = AB^2 + BC^2 = 576 + 49 = 625 \Rightarrow AC = 25 \text{ cm}$. (2)

SECTION C — Answer Key (3 marks each)

- Q26. $\text{LCM}(9, 12, 15) = 180 \text{ minutes} = 3 \text{ hours}$. They will next toll together at $8:00 + 3 \text{ hours} = \mathbf{11:00 \text{ a.m.}}$ (3)
- Q27. Sum of zeroes = $k/2 = 3 \Rightarrow k = 6$. (3)
- Q28. Let son's present age = x . Father = $3x$. After 12 years: $3x + 12 = 2(x + 12) \Rightarrow 3x + 12 = 2x + 24 \Rightarrow x = 12$. **Son = 12 years, Father = 36 years**. (3)
- Q29. $x + 1/x = 50/7 \Rightarrow 7x^2 - 50x + 7 = 0 \Rightarrow (7x - 1)(x - 7) = 0 \Rightarrow x = 1/7 \text{ or } 7$. So the number is **7 or 1/7**. (3)

Q30. Numbers between 100 and 500 divisible by 8: 104, 112, 120, ..., 496. $a = 104$, $d = 8$, $l = 496$. $n = (l - a)/d + 1 = (496 - 104)/8 + 1 = 50$. $\text{Sum} = (n/2)(a + l) = 25 \cdot 600 = \mathbf{15000}$. (3)

Q31. In $\triangle ABD$: $AB^2 = AD^2 + BD^2$. In $\triangle ACD$: $AC^2 = AD^2 + DC^2$. Subtract: $AB^2 - AC^2 = BD^2 - DC^2 \Rightarrow \mathbf{AB^2 - BD^2 = AC^2 - DC^2}$. ✓ (3)

SECTION D — Answer Key (5 marks each)

Q32. Proof. Suppose $\sqrt{7} = p/q$ (p, q coprime, $q \neq 0$). Then $7q^2 = p^2 \Rightarrow 7 \mid p^2 \Rightarrow 7 \mid p$. Let $p = 7k$. Then $7q^2 = 49k^2 \Rightarrow q^2 = 7k^2 \Rightarrow 7 \mid q^2 \Rightarrow 7 \mid q$. So 7 divides both p and q , contradicting that they are coprime. Hence $\sqrt{7}$ is irrational. (5)

Q33. Let speed of boat = u km/h, stream = v km/h. Downstream speed = $u + v$, upstream = $u - v$. Given: $44/(u + v) = 4 \Rightarrow u + v = 11$. And $28/(u - v) = 4 \Rightarrow u - v = 7$. Adding: $2u = 18 \Rightarrow \mathbf{u = 9 \text{ km/h}}$. So $\mathbf{v = 2 \text{ km/h}}$. (5)

Q34. $S_n = 7/2(2a + 6d) = 7(a + 3d) = 49 \Rightarrow a + 3d = 7 \dots$ (i). $S_{2n} = 17/2(2a + 16d) = 17(a + 8d) = 289 \Rightarrow a + 8d = 17 \dots$ (ii). Subtracting: $5d = 10 \Rightarrow d = 2$. Then $a = 7 - 6 = 1$. So $\mathbf{S_n = n/2(2 + (n-1) \cdot 2) = n^2}$. (5)

Q35. Converse of BPT: If a line divides any two sides of a triangle in the same ratio, then it is parallel to the third side.

Proof: In $\triangle ABC$, let line DE intersect AB at D and AC at E such that $AD/DB = AE/EC$. Suppose, if possible, DE is not parallel to BC . Then draw $DE' \parallel BC$ where E' is on AC . By BPT: $AD/DB = AE'/EC$. But given $AD/DB = AE/EC$, hence $AE'/EC = AE/EC \Rightarrow AE' = AE \Rightarrow E \equiv E'$. Therefore $DE \parallel BC$. ✓

Application — midpoint theorem: Let D and E be midpoints of AB and AC respectively in $\triangle ABC$. Then $AD/DB = 1 = AE/EC$. By converse of BPT, $DE \parallel BC$. ✓ (5)

SECTION E — Answer Key (4 marks each)

Q36. (i) AP: **20, 22, 24, 26, ...** ($a = 20$, $d = 2$). (1) (ii) $a_{10} = 20 + 24 \cdot 2 = \mathbf{68 \text{ seats}}$. (1) (iii) $S_{10} = (30/2)(2 \cdot 20 + 29 \cdot 2) = 15 \cdot (40 + 58) = 15 \cdot 98 = \mathbf{1470 \text{ seats}}$. (2)

Q37. (i) Let smaller side = x . Other side = $x + 7$. By Pythagoras: $x^2 + (x + 7)^2 = 13^2 \Rightarrow \mathbf{2x^2 + 14x - 120 = 0} \Rightarrow \mathbf{x^2 + 7x - 60 = 0}$. (2) (ii) $(x + 12)(x - 5) = 0 \Rightarrow x = 5$ (positive). Sides are **5 cm and 12 cm**. (2)

Q38. (i) **Yes — by AA similarity**; the sun's rays make equal angles, and both objects are vertical (90°). (1) (ii) $6/4 = h/28 \Rightarrow h = 6 \cdot 28/4 = \mathbf{42 \text{ m}}$. (2) (iii) $6/4 = 1.8/s \Rightarrow s = 1.8 \cdot 4/6 = \mathbf{1.2 \text{ m}}$. (1)

— End of Answer Key —